



S. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded
Department of Production Engineering
SUMMER TERM EXAMINATION

Academic Year: 2024-25

Class: Second Year	Subject: Thermal engineering/TE-II	Subject Code: PCC-PE206/207
Date: 23/06/2025	Time: 14:30 -17:30	Max. Marks : 100

Note

- i. Attempt all questions.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.

Q.1	Attempt the following :	M	CO	BT
a	Explain with neat sketch a vapour compression refrigeration cycle (VCRC).	10	CO1	BT1
b	Explain with neat sketch reverse carnot cycle with PV and TS diagram.	10	CO2	BT2
c	Define ; 1) Relative Humidity 2) Dew Point Temperature 3) Wet Bulb Temperature 4) Dry Bulb Temperature 5) Absolute Humidity. 6) Intensive Property 7) Extensive Property	10	CO1	BT1
d	Explain with neat sketch main components of IC engine.	10	CO3	BT3
Q.2	Solve the following ;			
a	Differentiate between petrol engine and diesel engine.	10	CO2	BT2
b	Explain with neat sketch Construction , Working of Sling Psychrometer.	10	CO3	BT2
Q.3	Solve:			
a	A Piston Cylinder device initially contains 0.4m^3 of air to be treated as an ideal gas at 100Kpa and 80°C . The air is now isothermally compressed to 0.1m^3 . The workdone during this process is	04	CO4	BT4
b	What are the types of system? And give its examples.	08	CO2	BT3
c	State and prove Gibb's phase rule.	08	CO3	BT4
Q.4	Explain with neat sketch Bell Coleman Cycle with PV and PS diagram.	10	CO2	BT4
Q.5	Explain with neat sketch any one Fire tube boiler.	10	CO1	BT1

=== BEST LUCK ===

Booms Taxonomy: BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create.



SGGS INSTITUTE OF ENGINEERING & TECHNOLOGY, VISHNUPURI, NANDED

Department of Production Engineering

Academic Year 2024-2025

Summer Term Examination Class: SY. (Prod)

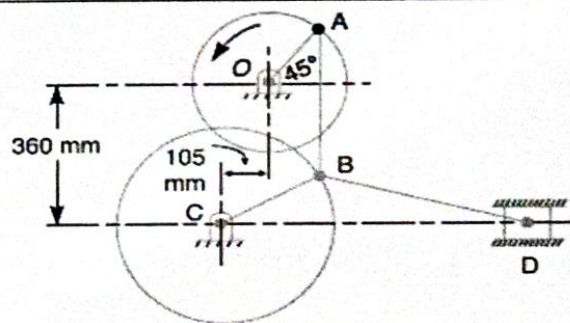
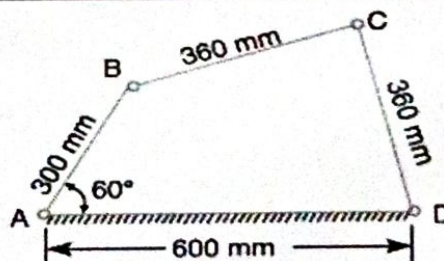
Dt: 20/06/25
14.30-17.30

Subject: Theory of machines PCC-PE206/PCC-PE204

Max Marks: 100

*CO: Course Outcomes.

Note: Solve any five questions from the following.

Note: Solve any live questions from the following.				
Q.1	Layout the profile of a cam so that the follower -Is moved outwards through 30mm during 180° of cam rotation with SHM. -Dwells for 20° of cam rotation, -Returns with uniform velocity during remaining 160° of cam rotation The base circle diameter of the cam is 28mm and the roller diameter 8mm. The axis of follower is offset by 6mm to the left. What will be the maximum velocity and acceleration of the follower during outstroke if the cam rotates at 1500rpm counter-clockwise?	20	CO7	
Q.2	The intercepted areas between the output torque curve and the mean resistance line of a turning moment diagram for a multicylinder engine, taken in order from one end are as follows: -35, +410, -285, +325, -335, +260, -365, +285, -260mm ² . The diagram has been drawn to a scale of 1mm= 70N-m and 1mm= 4.5° . The engine speed is 900rpm and the fluctuation in speed is not to exceed 2% of the mean speed. Find the mass and cross-section of the flywheel rim having 650mm mean diameter. The density of the material of flywheel may be taken as 7250kg/m ³ . The rim is rectangular with the width two times the thickness. Neglect effect of arms etc.	20	CO4	
Q.3	Each arm of a porter governor is 250mm long. The upper and lower arms are pivoted to links of 40mm and 50mm respectively from the axis of rotation. Each ball has a mass of 5kg and sleeve mass is 50kg. The force of friction on the sleeve mechanism is 40N. Determine the range of speed of the governor for extreme radii of rotation of 125mm and 150mm.	20	CO7	
Q.4	In the toggle mechanism shown in Fig.to right, the slider D is constrained to move on a horizontal path. The crank OA is rotating in the counter-clockwise direction at a speed of 180 r.p.m. The dimensions of the various links are: OA = 180 mm; CB = 240 mm; AB = 360 mm; and BD = 540 mm. For the given configuration, find 1. Velocity of slider D and angular velocity of links AB, CB, and BD.		20	CO3
Q.5	In a pin jointed four bar mechanism, as shown in Fig . , AB = 300 mm, BC = CD = 360 mm, and AD = 600 mm. The angle BAD = 60° . The crank AB rotates uniformly at 100 r.p.m. Locate all the instantaneous centres and find the angular velocity of the links BC and CD.		20	CO2
Q.6	a. Differentiate between lower kinematic pair and higher kinematic pair. And explain each type along with appropriate sketch. b. What are inversions of four bar mechanism? Explain in brief along with neat sketches. c. What is a machine? Giving example, differentiate between machine and mechanism. d. Explain with some example how degree of freedom is calculated.	05 05 05 05	CO1	



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
Summer-Term Semester Examination
Academic Year: 2024-25

Class: S.Y.	Subject: Engineering Metallurgy	Subject Code: PCC-PE204
Date: 19.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	What is Tempering? Explain the purpose of tempering and process of the tempering also list down various type of tempering and explain in detail anyone?	10	CO2	BT1
Q.2	a	What is TTT diagram explain in detail with advantages?	10	CO1	BT3
Q.3	a	Describe powder metallurgy, its processes and applications?	10	CO5	BT2
Q.4	a	What are composites materials and its application in manufacturing processes.	10	CO6	BT3
Q.5	a	Explain iron-iron carbide equilibrium diagrams	10	CO1	BT3
Q.6	a	Explain in detail difference between destructive testing and non-destructive testing?	10	CO6	BT2
Q.7	a	What is basis concept of open-heart Steel making process? Explain with neat sketch diagram?	10	CO2	BT3
Q.8	a	What is the different zones present in Blast-furnace which is used in pig-iron production. Explain briefly with neat sketch?	10	CO2	BT1
Q.9	a	Explain in details Magnetic particle test in details?	10	CO6	BT2
Q.10	a	Explain in details Ultrasonic test in details?	10	CO5	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Identify the methods of Manufacturing of steel and phase diagram. Interpret Iron -Iron carbide equilibrium diagram, time Temperature transformation (TTT) diagrams and their significance.
CO 2	Classify the cast irons and their applications and heat treatment processes.
CO 3	Select suitable heat-treatment process to achieve desired properties of steel by acquiring the fundamentals of heat treatment.
CO 4	Differentiate engineering nonferrous materials and its applications.
CO 5	Describe powder metallurgy, its processes and applications.
CO6	Choose the composite materials and its application in manufacturing processes.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
Summer Term Examination
Academic Year: 2024-25

Class: SY B.Tech.

Course: : Strength of Materials

Course Code: PCC-PE201

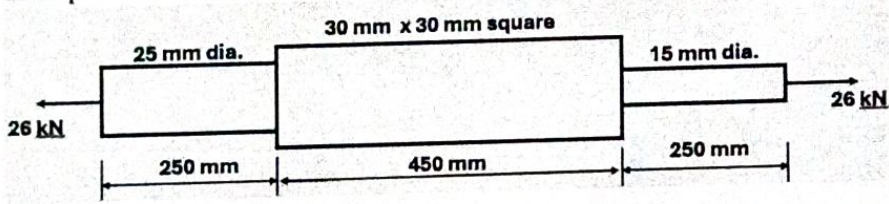
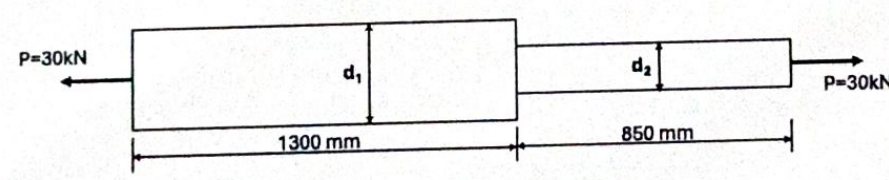
Date: 24.06.2025

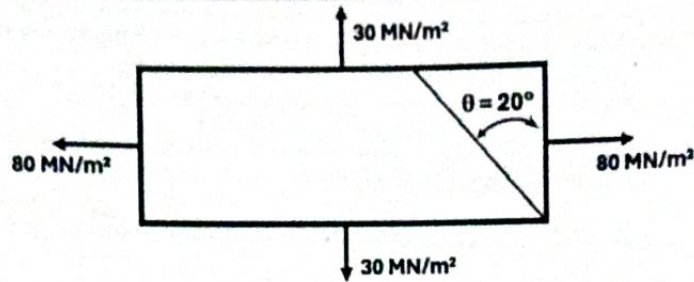
Time: 02.30 pm to 05.30 pm

Max. Marks: 100

Note:

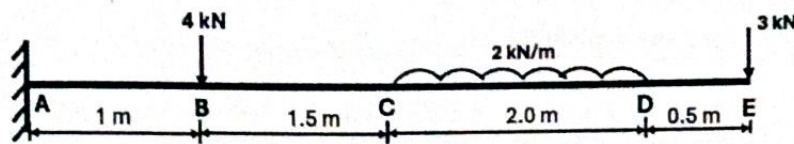
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- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Marks	CO	BT
Q.1	10	CO1	BT3
A bar ABCD 950 mm long is made up of three parts AB, BC, and CD of lengths 250mm, 450mm, and 250mm respectively. AB and CD are cylindrical having diameters 25 mm and 15 mm respectively. The BC is square cross section 30 mm x 30 mm. The rod is subjected to a pull of 26000 N. Find (i) The stresses in the three parts of the rod and (ii) the extension of the rod. Take $E = 3 \times 10^5 \text{ N/mm}^2$.			
			
Q.2	10	CO2	BT2
A steel bar 2150 mm long is 20 mm in diameter for half the length and 10 mm diameter for other half length. Find the extension of the bar when subjected to a tensile force of 30 kN.			
			
For a steel bar of uniform diameter having same length and volume what would be the extension when subjected to the same tensile force? Take $E = 3 \times 10^5 \text{ N/mm}^2$			
Q.3	10	CO2	BT3
The following data refer to a tensile test conducted on a mild steel bar. Diameter of the steel bar = 25 mm, Gauge length = 250 mm, Extension at a load of 200 kN = 0.50 mm, Load at elastic limit = 210 kN, Maximum load = 370 kN, Total extension = 60 mm, Diameter of the rod at failure = 20 mm. Calculate a) The Young's modulus (b) The stress at elastic limit, (c) percentage elongation (d) percentage decrease in area.			
Q.4	10	CO3	BT4
The principal stresses at a point across two perpendicular planes are 80 MN/m ² (tensile) and 30 MN/m ² (tensile). Find the Normal stress, Tangential stress, and Resultant stress in the magnitude and direction on a plane at 20° with the major principal plane. Use analytical and Mohr's circle method.			



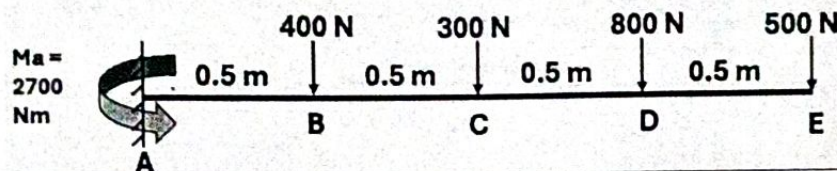
- Q.5 Calculate the shear force and bending moments at various cross sections of the beam i.e. at point A, B, C, D, E, and draw their variations along the length for the beam.

10 CO3 BT4



- Q.6 Calculate the shear force and bending moments at various cross sections of the beam as shown below and draw their variations along the length for the beam as shown below.

10 CO3 BT4



- Q.7 Explain the stress strain curve for tensile loading of brittle material with neat diagram. Explain the various points in detail.

10 CO2 BT2

- Q.8 Explain the following:
(i) Tensile stress and strain (ii) Young's Modulus (iii) Modulus of Rigidity (iv) Bulk or volume modulus of elasticity (v) Shear stress and strain.

10 CO1 BT1

- Q.9 A solid circular shaft transmits 75 kW power at 200 rpm. Calculate the shaft diameter if the twist in the shaft does not exceed 1 degree in 2 meters length of the shaft, and shear stress is limited to 50 MN/m². Take $C = 100 \text{ GN/m}^2$.

10 CO4 BT4

- Q.10 A circular bar made of cast iron resists a torque of 2.2 kNm acting in the transverse direction. If the allowable stresses in compression, tension, and shear are 100 MN/m², 35 MN/m², and 50 MN/m² respectively. Find the diameter of the bar and the angle of twist under the applied torque per meter length of the bar. Take $C = 40 \text{ GN/m}^2$.

10 CO5 BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

- On successful completion of this course, students should be able:
- CO 1 To differentiate the mechanical behaviour of engineering materials.
 - CO 2 To apply the theories of elastic failure.
 - CO 3 To determine the shear force and bending moment distribution of the beam and locate the point of contra flexure.
 - CO 4 To calculate stresses in beams for various boundary conditions.
 - CO 5 To determine the slope and deflection of the beam for various boundary conditions.